Sequences — problem 9.38

A. Engel — Problem-Solving Strategies · 1998

Three sequences x_n, y_n, z_n, $n = 1, 2, \ldots$ are defined as follows :

$$x_1 = 2,\ y_1 = 4,\ z_1 = \frac{6}{7}, \qquad x_{n+1} = \frac{2x_n}{x_n^2 - 1}, \quad y_{n+1} = \frac{2y_n}{y_n^2 - 1}, \quad z_{n+1} = \frac{2z_n}{z_n^2 - 1}.$$

(a) Show that this construction can be extended indefinitely.
(b) At some stage can we get $x_n + y_n + z_n = 0$ (ARO 1990) ?